Which Error Can the District Afford?

Unit 3 · Topic 3.8 · TODAY'S PROBLEM

A district checks an emergency-alert system. Let p be the proportion of alerts that arrive late. It tests $H_0 : p = 0.05$ versus $H_a : p > 0.05$, replacing the system if it rejects H_0. Both plans use $n = 200$. The Safety office wants to avoid keeping a system with a 10% late rate. The Budget office wants to avoid replacing a system with a 5% late rate. The offices have not agreed how to compare these costs.

Test plans ($n = 200$)

Plan	α	Power at $p = 0.10$
A	0.010	0.748
B	0.050	0.877

Dev: "Use A. Smaller α lowers both error risks."
Lin: "Use B. Its higher power has no cost when safety matters."

FIRST TAKE (5 min, on your sheet)

Which plan would you try first? Name the office you prioritized and one competing cost.

DOK 1 **(a)** Identify what the table's power at $p = 0.10$ means in this setting.

DOK 2 **(b)** Calculate the Type II error probability β for each plan. State each plan's Type I error probability.

DOK 3 **(c)** In 4–5 sentences, recommend a plan for one office using both error risks. Explain the consequence of each error and what Dev and Lin each overlook. State why the other office might choose differently.

Optional review: follow-along (scan the code)
Work (a)–(c) from the rules on your sheet. Turn in your sheet.